Week 9-2: Decision Trees

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Decision Tree

- Dataset

X_1	X_2	Y
M	Y	1
F	Y	1
M	N	0

- Rule Representation:
  - If $X_1 = F$ then $Y = 1$
  - If $X_1 = M$ and $X_2 = N$ then $Y = 0$

Example of data set

day	outlook	temp	humidity	wind	play
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

Entropy

- A measure of impurity, uncertainty, randomness, in a set of examples
- **Formula:**

$$H(X) = -\sum_{i=1}^{k} p(x_i) \log_2 p(x_i), \quad \text{where k is a number of classes.}$$

- **Example:**

$$H(play) = -\frac{9}{14} \log_2 \frac{9}{14} - \frac{5}{14} \log_2 \frac{5}{14} \approx 0.940$$

- Ref: `https://victorzhou.com/blog/information-gain/`

Information Gain

- How can we quantify the quality of a split?
- Higher Gain $\Rightarrow$ better split.

Definition:

$$\begin{aligned}\text{Gain}(S, A) &= H(S) - H(S|A) \\ &= H(S) - \sum_{v \in \text{Values}(A)} \frac{|S_v|}{|S|} H(S_v)\end{aligned}$$

- S: set of examples
- A: attribute
- Values(A): possible values of A
- S_v: subset of S where the attribute A has the specific value v
- Ref: `https://victorzhou.com/blog/information-gain/`

Example: Gain(S, Humidity)

$$
\begin{aligned}
\text{Gain}(S, \text{Humidity}) &= H(S) - \sum_{v \in \{\text{high,normal}\}} \frac{|S_v|}{|S|} H(S_v) \\
&= 0.94 - \Big[\frac{7}{14}\Big(-\frac{3}{7} log_2 \frac{3}{7} - \frac{4}{7} log_2 \frac{4}{7}\Big) + \\
&\qquad \frac{7}{14}\Big(-\frac{6}{7} log_2 \frac{6}{7} - \frac{1}{7} log_2 \frac{1}{7}\Big)\Big] \\
&= 0.151
\end{aligned}
$$

Example: Gain(S, Outlook)

$$\text{Gain}(S, \text{Outlook}) = H(S) - \sum_{v \in \{\text{Sunny}, \text{Overcast}, \text{Rain}\}} \frac{|S_v|}{|S|} H(S_v)$$

Example: Gain(S, Wind)

$$\text{Gain}(S, \text{Wind}) = H(S) - \sum_{v \in \{\text{Weak},\text{Strong}\}} \frac{|S_v|}{|S|} H(S_v)$$

Example: Gain(S, Temp)

$$\text{Gain}(S, \text{Temp}) = H(S) - \sum_{v \in \{\text{Hot},\text{Mild},\text{Cool}\}} \frac{|S_v|}{|S|} H(S_v)$$

Information Gain for each attribute

IG(S, outlook) = 0.248 (How to get this value?)

IG(S, humidity) = 0.151

IG(S, wind) = 0.048

IG(S, temp) = 0.029

Choose attribute with **highest IG** for a root of Tree.

ID3 algorithm (part 1)

```
def id3(examples, classification_attribute, attributes):
    create a root node for the tree
    if all examples are positive/yes:
        return root node with positive/yes label
    else if all examples are negative/no:
        return root node with negative/no label
    else if there are no attributes left:
        return root node with most popular
classification_attribute label
    else:
```

ID3 algorithm (part 2)

```
best_attribute = attribute from attributes that best
                 classifies examples
assign best_attribute to root node
for each value in best_attribute:
    add branch below root node for the value
    branch_examples = [examples that have that value
                       for best_attribute]
    if branch_examples is empty:
        add leaf node with most popular
            classification_attribute label
    else:

        add subtree id3(branch_examples,
                        classification_attribute,
                        attributes - best_attribute)
```

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Subset Example for Outlook (Sunny) with humidity

ID3([D1, D2, D8, D9, D11],
[N, N, N, Y, Y],
[Humidity, Wind, temp])

Compute:

$$IG(S_{\text{sunny}}, \text{Humidity}) = H(S_{\text{sunny}}) - \sum_{v \in \{\text{high,normal}\}} \frac{|S_v|}{|S|} H(S_v)$$

$$= -\frac{2}{5} log_2(\frac{2}{5}) - \frac{3}{5} log_2(\frac{3}{5})$$

$$- \left[\frac{3}{5}(0) + \frac{2}{5}(0)\right]$$

$$= 0.97$$

Subset Example for Outlook (Sunny) with temp

$$
\begin{aligned}
IG(S_{\text{sunny}}, \text{temp}) &= H(S_{\text{sunny}}) - \sum_{v \in \{\text{hot,mild,cool}\}} \frac{|S_v|}{|S|} H(S_v) \\
&= 0.97 \\
&\quad - \left[\tfrac{2}{5}(0) + \tfrac{2}{5}(1) + \tfrac{1}{5}(0)\right] \\
&= 0.57
\end{aligned}
$$

Subset Example for Outlook (Sunny) with wind

$$
\begin{aligned}
IG(S_{\text{sunny}}, \text{wind}) &= H(S_{\text{sunny}}) - \sum_{v \in \{\text{weak,strong}\}} \frac{|S_v|}{|S|} H(S_v) \\
&= 0.97 \\
&\quad - \left[\tfrac{3}{5}(0.92) + \tfrac{2}{5}(1)\right] \\
&= 0.018
\end{aligned}
$$

Decision Tree Overview

- ID3 $\rightarrow$ Not available in scikit-learn
- C4.5 $\rightarrow$ Handles continuous data
- CART (Classification and Regression Tree)
  - Binary tree
  - Uses Information Gain (IG), Gini Index (Impurity)
  - Used as splitting criterion

Splitting Criteria

1. **Information Gain** (based on Entropy)
2. **Gini Index** (Impurity)

Gini Index formula:

$$GI = 1 - \sum_{i=1}^{K} p_i^2 \quad \text{where} \quad p_i = \frac{\text{value of each class}}{\text{Total}} \quad \text{,and}$$

$$K \quad \text{is a number of classes.}$$

Example of data set

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Gini Index for Other Attributes

$$Gl(\text{outlook}) = \frac{5}{14}(0.48) + \frac{4}{14}(0) + \frac{5}{14}(0.48) = 0.342$$

$$Gl(\text{temperature}) = \frac{4}{14}(0.5) + \frac{6}{14}(0.44) + \frac{4}{14}(0.375) = 0.439$$

$$Gl(\text{humidity}) = \frac{7}{14}(0.48) + \frac{7}{14}(0.24) = 0.367$$

$$Gl(\text{wind}) = \frac{8}{14}(0.375) + \frac{6}{14}(0.5) = 0.428$$

$\therefore$ Pick the lowest value, i.e., outlook, as the root node.

Overfitting in Decision Trees

- High depth $\Rightarrow$ Too many nodes
- **Solution to overfit:**
  1. Limit the number of depths or iterations (e.g., ID3).
  2. Pruning (Post-pruning): adjust after training.
  3. Bagging VS Boosting.
  4. Ref1: `https://www.kaggle.com/code/prashant111/bagging-vs-boosting`
  5. Ref2: `https://www.kaggle.com/code/satishgunjal/ensemble-learning-bagging-boosting-stacking`

Pruning: Bottom-Up Walk

- Case 1: Replace internal node with a leaf.
- Case 2: Replace internal node with a subtree.

Bagging (Bootstrap Aggregation)

Algorithm 1: Bagging with Decision Trees

Input: Training dataset $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{n}$, number of estimators M

for $m = 1$ **to** M **do**

  Generate bootstrap sample $\mathcal{D}_m$ by sampling n instances from $\mathcal{D}$ with replacement;

  Train decision tree $h_m(x)$ on $\mathcal{D}_m$;

Output: Ensemble of trees $\{h_m(x)\}_{m=1}^{M}$

Boosting (ADABoost)

$\textsc{AdaBoost}(\mathcal{D}_n, \textsc{Base}(\cdot,\cdot), T)$

$\mathbf{w}^{(1)} \leftarrow (1/n, \ldots, 1/n)$ ▷ *initial weights*

for $t \leftarrow 1$ **to** T

  $h^{(t)} \leftarrow \textsc{Base}(\mathcal{D}_n, \mathbf{w}^{(t)})$ ▷ *base classifier*

  $\varepsilon^{(t)} \leftarrow \sum_{i=1}^{n} w_i^{(t)} \mathbb{I}\left\{h^{(t)}(\mathbf{x}_i) \neq y_i\right\}$ ▷ *weighted error of the base classifier*

  $\alpha^{(t)} \leftarrow \frac{1}{2} \ln\left(\frac{1-\varepsilon^{(t)}}{\varepsilon^{(t)}}\right)$ ▷ *coefficient of the base classifier*

  for $i \leftarrow 1$ **to** n ▷ *re-weighting the training points*

    if $h^{(t)}(\mathbf{x}_i) \neq y_i$ **then** ▷ *error*

      $w_i^{(t+1)} \leftarrow \frac{w_i^{(t)}}{2\varepsilon^{(t)}}$ ▷ *weight increases*

    else ▷ *correct classification*

      $w_i^{(t+1)} \leftarrow \frac{w_i^{(t)}}{2\left(1-\varepsilon^{(t)}\right)}$ ▷ *weight decreases*

return $f^{(T)}(\cdot) = \sum_{t=1}^{T} \alpha^{(t)} h^{(t)}(\cdot)$ ▷ *weighted "vote" of base classifiers*

Figure 2.1: The pseudocode of ADABOOST. $\mathcal{D}_n = \{(\mathbf{x}_1, y_1), \ldots, (\mathbf{x}_n, y_n)\}$ is the training set, $\textsc{Base}(\cdot,\cdot)$ is the base learner, and T is the number of iterations.

Ref: Kégl, B. (2009). Introduction to adaboost.

Reference

- Song,Y.Y.,and Ying,L.U.(2015). Decision tree methods: applications for classification and prediction.
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